# Yunfeng Zhang

CURRICULUM VITAE

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## Research Interests

**Areas:** Harmonic analysis, representation theory, analytic number theory, and dispersive partial differential equations.

**Topics:** Bounds for characters of compact Lie groups and for spherical functions and joint eigenfunctions of invariant differential operators on symmetric spaces; bounds for Laplace eigenfunctions on manifolds; bounds for Hecke eigenforms; Strichartz estimates for the Schrödinger equation; and well-posedness and long-time behavior of nonlinear dispersive equations.

## Education

**Ph.D. in Mathematics, University of California, Los Angeles** 2012–2018
Advisors: Rowan Killip and Monica Visan

**B.S. in Mathematics, Tsinghua University** 2008–2012

## Academic Appointments

**University of Mississippi** 2026–present
Visiting Assistant Professor

**University of Cincinnati** 2024–2026
Visiting Assistant Professor

**Peking University** 2021–2024
TAL Assistant Professor

**University of Connecticut** 2018–2021
Assistant Research Professor

## Honors and Awards

UCLA Mathematics Graduate Research Presentation Prize 2018

Tsinghua University Outstanding Graduate Award 2012

Fellowship in the Tsinghua Xuetang Mathematics Program 2009–2012

## Grants

**Co-I (30% effort), National Key Research and Development Program of China** 2022–2027
Geometry and Analysis of Homogeneous Spaces
Total award: CNY 3,000,000

**PI, Fundamental Research Funds for the Central Universities, Peking University** 2021–2023
Analysis on Lie Groups
Total award: CNY 200,000

## Journal Publications

10. **Pointwise character bounds for $SU(3)$**
*Bulletin of the London Mathematical Society* 58 (2026), No. 10, Article ID e70519. arXiv:2604.17589

9. **Growth of Fourier–Lebesgue norms for mKdV**
With Saikatul Haque, Rowan Killip and Monica Visan.
*Partial Differential Equations and Applications* 7 (2026), No. 4, Article No. 31. arXiv:2511.17471

8. **Restriction of eigenfunctions on products of spheres to submanifolds of maximal flats**
*Proceedings of the American Mathematical Society* 154 (2026), No. 9, 3921–3931. arXiv:2511.14615

7. **Local well-posedness for nonlinear Schrödinger equations on compact product manifolds**
*Journal of Differential Equations* 467 (2026), Article ID 114288. arXiv:2503.09442

6. **Bounds of restriction of characters to submanifolds**
*Mathematische Zeitschrift* 312 (2026), No. 1, Article No. 13. arXiv:2402.03178

5. **Global well-posedness and equicontinuity for mKdV in modulation spaces**
With Saikatul Haque, Rowan Killip and Monica Visan.
*Pure and Applied Analysis* 7 (2025), No. 3, 615–637. arXiv:2411.05300

4. **On Fourier restriction type problems on compact Lie groups**
*Indiana University Mathematics Journal* 72 (2023), No. 6, 2631–2699. arXiv:2005.11451

3. **Schrödinger equations on compact globally symmetric spaces**
*Journal of Geometric Analysis* 31 (2021), No. 11, 10778–10819. arXiv:2005.00429

2. **Strichartz estimates for the Schrödinger equation on products of odd-dimensional spheres**
*Nonlinear Analysis* 199 (2020), Article ID 112052. arXiv:2301.02823

1. **Strichartz estimates for the Schrödinger flow on compact Lie groups**
*Analysis & PDE* 13 (2020), No. 4, 1173–1219. arXiv:1703.07548

## Preprints

3. **Rootwise estimates for Weyl alternants**
With Xiaocheng Li. arXiv:2609.32214

2. **Sharp bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy critical NLS**
With Yangkendi Deng and Zehua Zhao. arXiv:2509.09565

1. **Algebraic and analytic properties of invariant differential operators on a homogeneous space of complexity one**
With Hanlong Fang and Xiaocheng Li. arXiv:2301.00529

## Book Chapters

1. **Analysis on compact symmetric spaces: eigenfunctions and nonlinear Schrödinger equations**
In: *Trends in Mathematics, Research Perspectives Ghent Analysis and PDE Center 3* (2024), 235–240.

## Talks

* *denotes a contributed talk; unmarked talks are invited.*

**2026**

Oct. **Rootwise estimates for Weyl alternants***
Palmetto Number Theory Series (PANTS) XLI, Georgia Institute of Technology

Apr. **Bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy-critical NLS**
Applied Analysis Seminar, Louisiana State University

Apr. **Pointwise character bounds: the case of** $\mathrm{SU}(3)$
Analysis Seminar, University of Cincinnati

Mar. **Bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy-critical NLS**[*]
The Fifteenth Ohio River Analysis Meeting, University of Kentucky

**2025**

Nov. **Well-posedness of the energy-critical NLS on** $\mathbb{R} \times \mathbb{S}^3$
Beijing Institute of Technology

Oct. **Well-posedness of the energy-critical NLS on** $\mathbb{R} \times \mathbb{S}^3$
Analysis Seminar, Bielefeld University

Jun. **On NLS posed on** $\mathbb{R} \times \mathbb{S}^3$
Workshop on Dispersive PDEs and Control Theory, Beijing Institute of Technology

May **Bounds of restriction of characters to submanifolds**
Tsinghua University

Apr. **The modified KdV equation beyond Sobolev spaces**
Analysis Seminar, University of Cincinnati

Mar. **Bounds of restriction of characters to submanifolds**
AMS Special Session on Recent Trends in Harmonic Analysis and PDE, University of Kansas

Mar. **Multi-linear multi-parameter eigenfunction bounds and NLS on compact manifolds**
Beijing Institute of Technology

**2024**

Dec. **On the modified KdV equation in modulation spaces**
Joint Meeting of the NZMS, AustMS and AMS: Special Session on Harmonic Analysis and Hamiltonian PDE, University of Auckland

Oct. **Semiclassical fun with** $\mathrm{SU}(3)$[*]
Prairie Analysis Seminar 2024, University of Kansas

Sep. **Semiclassical fun with** $\mathrm{SU}(3)$
Analysis Seminar, University of Cincinnati

Jun. **Bounds of restriction of characters to submanifolds**
Analysis Seminar, Southern University of Science and Technology

Jun. **The modified KdV in modulation spaces: conservation laws and equicontinuity of solutions**
Beijing Institute of Technology

May **Bounds of restriction of characters to submanifolds**
Analysis Seminar, University of Wisconsin–Madison

Jan. **Bounds of restriction of characters to submanifolds**
Beijing Institute of Technology

**2023**

Dec. **Harmonic analysis on compact symmetric spaces**[*]
Global Young Scholars Forum, Beijing Normal University

Dec. $L^p$ **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Young Scholars Forum, ShanghaiTech University

Dec. $L^p$ **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Young Mathematician Forum, Shanghai Jiao Tong University

Dec. **Harmonic analysis on compact symmetric spaces**[*]
Vision Forum for International Young Scholars, Beihang University

Nov. $L^p$ **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Global Forum for Young Mathematicians, Southern University of Science and Technology

Nov. $L^p$ **norms of Laplacian eigenfunctions on compact Lie groups**[*]
Teli Forum for International Young Scholars, Beijing Institute of Technology

Sep. **Discrete Fourier restriction and the Kloosterman circle method**
Colloquium, Huaibei Normal University

Sep. **Fourier restriction type problems on compact Lie groups**
Beijing Institute of Technology

**2021**

Nov. **Nonlinear Schrödinger equation on compact symmetric spaces**
Methusalem Junior Analysis & PDE Seminar, Ghent University

Oct. **Fourier restriction bounds on compact symmetric spaces**
Conference on Harmonic Analysis and Symmetric Spaces, University of Wisconsin–Madison

Apr. **Strichartz estimate for the Schrödinger equation on compact globally symmetric spaces**
Oberseminar Analysis, Bielefeld University

Feb. **Schrödinger equations on compact globally symmetric spaces**
Weekly Seminar on Geometric and Functional Inequalities and Applications, University of Connecticut

**2020**

Sep. **Size of Laplacian eigenfunctions on compact symmetric spaces**
AMS Special Session on Geometric Inequalities and Nonlinear PDEs, University of Texas at El Paso

Mar. **Strichartz estimates for the Schrödinger equation on compact symmetric spaces**
AMS Special Session on Analysis on Homogeneous Spaces, Tufts University (Cancelled due to COVID)

## Teaching

### AS INSTRUCTOR

**University of Mississippi**

| | |
|---|---|
| Elementary Differential Equations (two classes) | Spring 2026 |
| Elementary Differential Equations | Fall 2026 |
| Unified Calculus and Analytic Geometry III — honors section | Fall 2026 |
| Unified Calculus and Analytic Geometry III — standard section | Fall 2026 |

**University of Cincinnati**

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|---|---|
| Linear Algebra | Spring 2026 |
| Calculus II (two sections) | Fall 2025 |
| Pre Calculus | Fall 2025 |
| Calculus I | Spring 2025 |
| Applied Calculus I | Spring 2025 |
| College Algebra (two sections) | Fall 2024 |

**Peking University**

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|---|---|
| Linear Algebra B (“B” stands for “for the Physical Sciences”) | Fall 2023 |
| Linear Algebra B | Fall 2022 |
| Advanced Mathematics B (i.e. Calculus for the Physical Sciences) | Fall 2021 |

**University of Connecticut**

| | |
|---|---|
| Partial Differential Equations (two classes) | Spring 2021 |
| Partial Differential Equations (two classes) | Fall 2020 |
| Axiomatic Geometry (two classes) | Spring 2020 |
| Introduction to Complex Variables (two classes) | Fall 2019 |
| Partial Differential Equations (two classes) | Spring 2019 |
| Honors Calculus II | Fall 2018 |
| Honors Multivariable Calculus | Fall 2018 |

**University of California, Los Angeles**

Calculus for Life Sciences Students II — Summer 2017

### AS TEACHING ASSISTANT

**University of California, Los Angeles**

| | |
|---|---|
| Probability Theory II | Spring 2018, Spring 2017, Winter 2017, Winter 2016 |
| Algebra for Applications | Winter 2018 |
| Analysis I | Fall 2017, Winter 2016, Fall 2015 |
| Probability Theory I | Winter 2017, Winter 2015 |
| Differential and Integral Calculus | Fall 2016 |
| Linear & Nonlinear Systems of Differential Equations | Fall 2015, Spring 2015, Winter 2014 |
| Mathematical Game Theory | Summer 2015 |
| Partial Differential Equations | Spring 2015 |
| Discrete Structures | Winter 2015 |
| Precalculus | Fall 2014, Fall 2012 |
| Calculus for Life Sciences Students I | Fall 2014 |
| Linear Algebra I | Summer 2014 |
| Differential Geometry II | Spring 2014 |
| Ordinary Differential Equations | Spring 2014, Winter 2014 |
| Integration and Infinite Series | Fall 2013 |
| Complex Analysis for Applications | Spring 2013 |
| Differential Equations | Winter 2013 |

## Mentoring

**An Nguyen**, undergraduate student at the University of Cincinnati — 2025–2026

Guided his reading of John Stillwell's *Naive Lie Theory* and proposed a research problem on nonlinear Schrödinger equations on products of spheres involving representations of $\mathrm{SU}(2)$.

## Service and Outreach

**Co-organizer** of the minisymposium "Recent Developments in Applied Harmonic Analysis" at the 11th Annual Meeting of the SIAM Central States Section, Jackson State University (2026)

**Judge** for the 40th (2025) and 41st (2026) Annual UC Math Bowl — a mathematics competition for middle and high school students hosted by the University of Cincinnati

**Co-organizer** of the Analysis and Probability Seminar at the University of Connecticut, Fall 2020 and Spring 2021

**Reviewer** for Mathematical Reviews and zbMATH Open

**Referee** for

*Beijing Journal of Pure and Applied Mathematics*
*Bulletin of the London Mathematical Society*
*Communications on Pure and Applied Analysis*
*Journal of Differential Equations*
*Journal of Functional Analysis*
*Journal of Pseudo-Differential Operators and Applications*
*Pacific Journal of Mathematics*
*Selecta Mathematica* (quick opinion)
*Transactions of the American Mathematical Society*